

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Owned an AI initiative end-to-end — from use-case identification through PoC to production adoption: designed and shipped a self-service analytics agent over 67 BigQuery tables (sub-agent orchestration, 8 deterministic SQL rules, 23-case golden-query evaluation suite, human-in-the-loop validation). Partnered with Product, Data Science, and business stakeholders; approved by Sr. Director and product leadership; currently the only AI skill in active use by business teams on Data Ventures.
- Built and shipped to production an autonomous Jira ticket-resolution agent with agentic RAG (6-gate triage, MCP-driven context gathering, FAISS + BM25 retrieval over 11,000 historical SQL queries, self-healing execution); closed 400+ data requests across 8 categories over 15 months and cut routine turnaround from 30–60 minutes to under 10 minutes — measurable impact, not a recommendation deck.
- Built a registry-governed semantic layer over BigQuery with deterministic validation, value canonicalization, guardrails, and a human review gate; designed for sensitive, governed data with output validation so the same question returns the same answer every time.
- Built a hybrid orchestrator that lets Product self-serve requests they previously filed as tickets; routes questions through a semantic-layer skill and a context-engineered agent, reasons about disagreements, and refines definitions over time — driving sustained adoption, not a one-off project.
- Translated ambiguous business problems into AI use cases for non-technical stakeholders and facilitated their adoption; delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- Delivered a RAG-to-proposal generator for an enterprise consulting client, deployed on AWS EC2; scoped the use case with the client and owned delivery end-to-end; cut proposal time ~30%; **Best in Show — UChicago MSADS Capstone.**

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; designed controlled A/B tests that persuaded 15% to donate to an opposing cause.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); partnered with district stakeholders; retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit front-end + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

AI delivery & solutioning: use-case identification, PoC delivery, end-to-end ownership (scope → adoption), stakeholder/business translation, client-facing communication, cross-functional leadership, mentoring

AI / LLM products: agentic workflows & tool use, AI agents with reasoning/planning, agent orchestration, RAG & vector search (embeddings + FAISS + BM25 + RRF), MCP server design, prompt engineering, evals & testing (golden sets, human-in-the-loop), guardrails / output validation, responsible-AI & PII awareness, multi-model routing

Backend / production: Python, FastAPI, Cloud Run, REST APIs, LLM APIs & integration, CI/CD (pytest), headless cron agents, decision logging / observability

Data / SQL: BigQuery, dbt, SQL (complex query design + optimization), vector databases, PySpark, Airflow, Databricks, MySQL

LLMs / cloud AI: Claude (Opus / Sonnet / Haiku), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

Viz: Power BI (DAX, XMLA TMSL), Tableau, Streamlit, Matplotlib

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL